How to efficiently offer support/resources to train students who might start studying SDM approaches but need a lot of guidance to get going?

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So, you want to learn how to create and utilize SDMs~~.~~? First, what is an SDM? An SDM, or a species distribution model is a now~~- a~~ common method in the fields of biogeography, ecology, and conservation to predict species distribution, movement, and/or abundance across a landscape. You've probably heard several terms that all fall under the umbrella of species distribution modelling, including habitat models, climate envelopes, ecological niche models (ENMs), resource selection functions (RSFs), bioclimatic models, connectivity models, and range maps. SDMs use algorithms to fit mathematical functions that describe species presence/absence across environmental space and the relationship to each environmental variable and then generate~~s~~ predictive maps of species distribution in geographical space, aka predictive forecasting (Figure 1). In some cases, SDMs are used to explain or understand how environmental conditions influence the occurrence or abundance of a species.

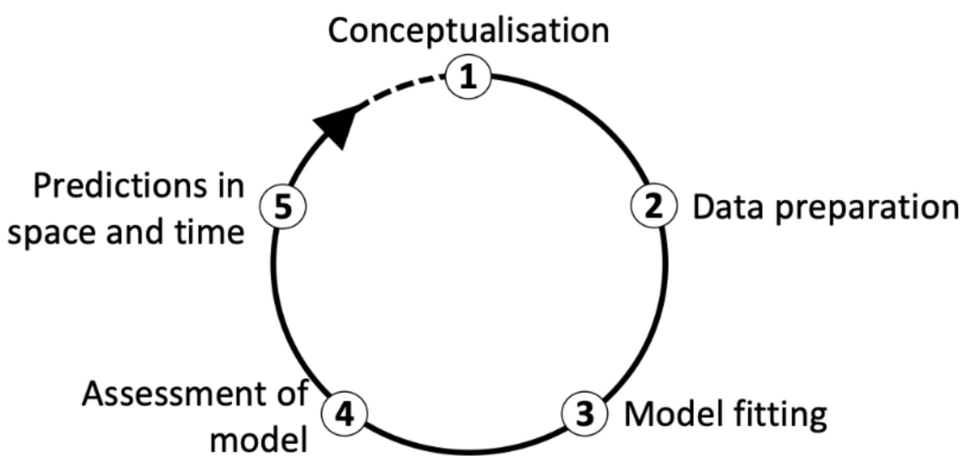

Figure 1: The main modelling cycle in species distribution modelling (Zurell, 2020).

There are two main data components to an SDM: (1) species data and (2) environmental data. Species data can comprise presence-only data (just where that species has been observed), presence-absence data (known presence and absence locations based on surveys), abundance-absence data (which includes counts), or occupancy data (repeated site surveys~~known repeat use of sites~~). ~~can use either occurrence only data or presence/absence data, but Rr~~Regardless of the data type you need to have coordinates of the locations where the species of interest occurred (or *didn't* occur). These can be archived~~pre-collected~~ species location data, such as from museum records or citizen science initiatives (e.g., ebird, iNaturalist, FrogWatch, etc.), or a new dataset you collect in the field~~self-collected samples~~. Environmental data is both the biotic and abiotic conditions that describe ~~the conditions of~~ where a species is present or absent. Now, there are dozens of variables to choose from to incorporate into your SDM. The variables you choose largely depend on the scale of your model and the completeness of data available.

But how much data do you need? Where do you get this data? What do I do? Unfortunately, this is not the place to learn all the ins and outs for SDMs (Once you have the basics, this FAQ page is incredibly beneficial to answer some of your more nuanced questions. Or feel free to contact some of our experts for more information or specific questions to help you out! (contact info or contact page?).

~~.~~As you explore SDMs, tThere are HUNDREDS of resources out there and available to help you meet your needs with learning how to utilize this powerful data tool to your advantage. However, we have put together a shortlist of~~There are severa~~l platforms and base programs to choose from (Table 1). Each of them has their benefits, caveats, and data requirements. Table 1, a non-exhaustive list, describes some commonly used platforms, required data, and links to further tutorials or readings to help you on your way!

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Method	**Model Name/Software**	**Type**	**Species Data Type**	**Links for Further Learning**
Climatic envelope	BIOCLIM	Probability	Presence-only	R tutorial; Booth, 2019
Gower metric	DOMAIN	Probability	Presence-only	Carpenter et al., 1993; R tutorial
Ecological niche factor analysis (ENFA)	BIOMAPPER	Suitability	Presence/background	BioMapper tutorial
Maximum entropy	MAXENT	Suitability	Presence/background	YouTube tutorial; Maxent step by step
Genetic algorithm	GARP	Suitability	Presence/pseudo-absence	Haase et al., 2021; Tutorial
Regression: generalized linear model (GLM) and generalized additive model (GAM)	GRASP	Probability	Presence/absence	Tutorial; Guisan et al, 2002.
Artificial neural network (ANN)	SPECIES, MATLAB	Probability	Presence/absence	Introduction; YouTube tutorial
Classification and regression trees (CART)	BIOMOD	Probability	Presence/absence	Tutorial
Multivariate adaptive regression splines (MARS)	MARS (with R)	Probability	Presence/absence	Introduction, Elith and Leathwick, 2007; R tutorial
Favourability function (FF)	FF (with R)	Probability	Presence/absence	Introduction; R tutorial
Random forest (RF)	RF (with R)	Probability	Presence/background	Valavi et al., 2021.
Boosted regression trees (BRT)	BRT (with R)	Probability	Presence/absence	R tutorial; Elith et al., 2008
Bayesian species distribution models	spOccupancy (in R)	Probability	Presence/absence	Single-Species, Multi-Species, and Integrated

				Spatial Occupancy Models • spOccupancy (jeffdoser.com)

Other Helpful Links:

Choosing a model method.

Video: Introduction to SDM.